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Sable Harbor Financial Build Readiness Report

Executive summary

Conclusion: Sable Harbor is not ready for canonical financial statements, but it is ready for a disciplined financial-model rebuild. The current `main` branch now contains enough locked institutional, portfolio, governance, financing, and operating logic to design a defensible accounting architecture. It does **not** contain enough quantitative canon to publish a trustworthy 2026 income statement, balance sheet, cash-flow statement, business-line P&Ls, or enterprise workforce census. Current `main` explicitly leaves exact enterprise headcount, revenue, product P&Ls, detailed legal mechanics, and several operating-business economics open.

The most important finding is that **the financial architecture in PR #9 is materially underscaled relative to September 3 canon**. PR #9 itself correctly labels its quantitative baseline as reversible and noncanonical, and the PR remains review-blocked and unaccepted. Its working case of 708 employees, $178.6 million of FY2026 revenue, $111 million of gross debt, and specific Red Wash/ARU economics is therefore useful modeling scaffolding—not company truth. Repository administration likewise records that PR #9 remains review-blocked and that current `main` must prevail over older branch assumptions.

The biggest new scale constraint is **J2**. September 3 canon establishes a 237-billet J2 establishment: 78 Contact, 42 Judgment, 24 Orientation, 30 Junction Advisory Group, 35 Education, and 28 headquarters billets. That is a design establishment rather than an assertion that all 237 seats are occupied, but it is nevertheless a locked institutional capacity requirement. At full establishment, J2 alone equals **33.5% of PR #9's entire 708-person modeled enterprise census**. If J2 were additive to that census, the combined population would be 945 and J2 would still represent 25.1%. A company in which a substantially occupied J2 represents a more sustainable 10–15% of total employment mechanically needs roughly **1,580–2,370 employees at full establishment**, or a lower J2 occupancy assumption that must be explicitly decided.

I found no controlling evidence for a **47,000-person workforce**. Current canon says the exact enterprise total remains open. Economically, 47,000 people would imply an enterprise radically larger than the story presently supports. In March 2026, BLS reported employer compensation of $75.15 per hour in professional/scientific/technical services, $83.81 in information, and $49.04 in transportation and warehousing; private-industry compensation at the 90th wage percentile averaged $89.70 per hour. [1] Given Sable Harbor's locked doctrine of strong cash compensation, serious professional depth, meaningful benefits, learning time as paid work, and long-duration employee participation, ordinary low-cost labor assumptions would be inappropriate. A 47,000-person scenario at $150,000–$225,000 of compensation-plus-role-support cost per employee implies **$7.1–$10.6 billion of annual workforce cost** and, if workforce cost were 40–50% of revenue, roughly **$14–$26 billion of annual revenue**. That is a scenario calculation, not canon, and there is no evidence of a company of that magnitude in the present history.

A better financial-build planning envelope is approximately **1,000–1,700 employees initially, with an economically healthier mature architecture around 1,500+ if J2 is substantially occupied**. This is a **SUPPORTED ESTIMATE**, not a census. It arises from the actual portfolio now in canon—Foundry Field, Atlas

Meridian, Willow, Pale Sun/Red Wash, Cradle, ARU/BS&T, Advisory, ESS, Internal Audit, and J2—rather than from preserving the obsolete 426-person software-company shell. The historical 426-FTE model was explicitly provisional and is now described by current `main` as historical/noncontrolling.

The preliminary revenue requirement is consequently far above PR #9's $178.6 million baseline. A company carrying the current institutional architecture plausibly needs **$450–$600 million merely to be economically sustainable**, approximately **$650–$900 million for a healthy operating model**, and roughly **$900 million–$1.2 billion for a high-performance model with material reinvestment capacity**. These are **SUPPORTED ESTIMATE / SCENARIO** ranges, not claimed history. The healthy-case midpoint is around **$700 million of annual revenue**, 52–60% consolidated gross margin, 10–15% EBITDA margin, positive operating cash generation, and sufficient free cash flow to maintain a software platform, a laboratory, an underground mine, logistics infrastructure, and a highly developed institutional center without relying perpetually on external financing.

The financing history is stronger than PR #9's inherited calibration suggests. Current locked governance records approximately **$48 million** of 2021 Harrison Vale financing and approximately **$135 million** of 2022 Wolf Ridge financing at approximately a **$900 million post-money valuation**, with no founder secondary in the latter round and a working post-round ownership split of approximately **66.5% insiders / 33.5% outside capital**. Those two locked rounds alone total approximately **$183 million**, already exceeding PR #9's inherited $172.5 million “equity capital raised through 2024” calibration. The Red Wash and ARU acquisition prices, acquisition financing, current debt, cash, leases, ARO, and subsequent ownership changes remain open.

The correct next move is therefore not to tweak PR #9's financial statements. It is to replace its company assumptions while preserving the useful accounting, provenance, run-isolation, validation, and export machinery.

Scope, authority and methodology

This review treats the September 2–3 `main`-branch canon as controlling. The repository README identifies the v0.3 lore canon, decision register, September 3 addendum, headquarters closeout, governance records, and related controlled documents as the current authoritative layer; it specifically warns that the old `architecture/corporate-operating-model-v0.1` branch is historical/noncontrolling. The v0.3 decision framework itself distinguishes locked facts from provisional direction and intentionally open facts; its quantitative operating-model decisions remain separate from lore canon.

This report therefore uses the following evidence discipline:

State	Treatment in this report
LOCKED	Controlling company fact; financial work must preserve it.
DERIVED	Arithmetic consequence of locked or explicitly identified assumptions.
SUPPORTED ESTIMATE	Research-grounded planning range appropriate for model design but not canon.

State	Treatment in this report
PROVISIONAL ASSUMPTION	Existing internal proposal that remains reversible.
SCENARIO	Deliberate what-if value used to test economics.
OPEN	Canon intentionally does not supply an answer.
CONFLICT	Two sources cannot simultaneously control; current `main` wins.
SUPERSEDED	Historical value preserved for provenance but unusable as current company truth.

External benchmarking is deliberately secondary to canon. It is used only to test economic scale and reasonableness. For labor, the principal source is the U.S. Bureau of Labor Statistics. For software and industrial economics, primary SEC filings and the Surface Transportation Board are used as bounding references rather than direct comparables. BLS's March 2026 data show particularly high employer-compensation costs in the professional/technical and information sectors, which is consistent with treating Sable Harbor's software, AI, advanced engineering, J2, legal, finance, and technical-specialist populations as expensive institutional capital rather than interchangeable headcount. [1]

Software comparisons demonstrate how wide the productivity range can be. Bentley Systems reported approximately $1.50 billion of 2025 revenue and roughly 5,800 FTE, while its subscription-heavy model produced gross margins above 80%; Palantir reported approximately $4.48 billion of 2025 revenue with 4,429 employees and similarly high gross margin. These are useful outer bounds, not claims that Foundry or Atlas can immediately achieve their economics. Sable Harbor mixes product engineering with implementations, mining expertise, physical operations, fieldwork, and institutional functions, so consolidated productivity should be materially lower than a high-performing pure software/AI company. [2]

Professional services provide a different constraint. FTI Consulting's 2025 Economic Consulting segment reported average billable rates around $583 per hour but only 59% utilization, while its Technology segment reported a 33.1% gross margin and 12.1% adjusted segment EBITDA margin. That illustrates why premium expert labor can support high rates without producing software-like margins. [3]

Physical operations impose yet another economics regime. The Surface Transportation Board's 2025 revenue threshold for Class II railroads is approximately $49.1 million; a rail operation below that threshold would remain Class III for regulatory revenue classification purposes. [4] Union Pacific, at the other extreme, reported $24.51 billion of 2025 operating revenue and a 59.8% operating ratio, illustrating the economics possible at enormous network scale but not providing an appropriate margin assumption for a small Sable Harbor-controlled railway. [5]

Uranium economics likewise cannot be reverse-engineered from employee counts. Recent U.S. producers show that revenue depends on pounds delivered, contract timing, realized price, inventory accumulation, and processing schedule. enCore reported 270,000 pounds of first-quarter 2026 uranium sales at an average $67.78 per pound, while Energy Fuels' 2025 reporting showed production and sale timing can differ materially as product is accumulated for later contracts. [6] That reinforces the canonically correct position

for Red Wash: production, reserves, workforce, economics, permits, closure costs, and mine plan must remain open until explicitly modeled.

Workforce census and cost architecture

The key distinction is between a **census** and a **planning envelope**. Sable Harbor currently has no locked enterprise census. The most concrete current workforce number is J2's 237-billet establishment, and even that is explicitly a design establishment rather than a mandate to fill every billet.

The earlier corporate-operating-model branch proposed 426 FTE and 34 long-term contractors, including 118 Foundry product/engineering staff, 52 Atlas personnel, 82 delivery personnel, 46 customer operations/SRE personnel, 27 Emberline staff, 12 Pale Sun personnel, 38 go-to-market staff, and small corporate functions. That document explicitly called every new numerical value provisional. Current `main` has since added or hardened a substantially larger institutional and physical portfolio, and the README now expressly says exact enterprise-wide 2026 headcount remains open.

PR #9 attempted to bridge the gap by treating the 426-person model as a CoreCo calibration and incrementally adding acquired industrial businesses. Its proposed 708-person census comprises approximately 431 CoreCo employees, 126 at Red Wash, 132 at ARU/BS&T, 12 at Cradle, and 7 dedicated to Advisory, plus 61 contractors. That is internally coherent as a modeling hypothesis but not reconciled to the September 3 J2 and headquarters architecture.

Reconciled readiness census

Institution / business	Controlling quantitative fact	Readiness planning envelope	State	Assessment
Foundry Field	Exact 2026 count OPEN	350–550	**SUPPORTED ESTIMATE**	Must encompass core product engineering plus substantial implementation, customer operations, field integration, and attributable commercial capacity.
Atlas Meridian	Exact count OPEN	60–100	**SUPPORTED ESTIMATE**	Controlled commercialization still requires applied science, product, model engineering, validation, mining expertise, safety/evaluation, and customer teams.
Willow	Exact count OPEN	35–65	**SUPPORTED ESTIMATE**	Pittsburgh experimental institution, field systems, prototypes, research, and Sacramento institutional seam.

Institution / business	Controlling quantitative fact	Readiness planning envelope	State	Assessment
Pale Sun / Red Wash	Workforce explicitly OPEN	100–170	**SUPPORTED ESTIMATE**	Underground mine, mill, maintenance, geology, environmental, safety, planning, supervision, and commercial/ accounting support; must ultimately be driver-built.
Cradle	Founding team is four; current count OPEN	10–25	**SUPPORTED ESTIMATE**	Small recovery-development organization rather than mine owner.
ARU / BS&T	Workforce explicitly OPEN	100–220	**SUPPORTED ESTIMATE**	Range remains intentionally wide until routes, terminals, fleet, train frequency, maintenance model, and customers are defined.
Advisory	Structure and P&L OPEN	10–40	**SUPPORTED ESTIMATE**	Emerging high-skill engagement practice; 2026 should remain comparatively small.
ESS, executive center, Internal Audit	Functions locked; counts OPEN	160–285	**SUPPORTED ESTIMATE**	Current canon demands materially more corporate capability than the 426-person model allowed.
J2	237-billet design establishment	170–237 occupied planning case	**SCENARIO**	Occupancy must not be confused with establishment.
Enterprise	Exact total OPEN	**~1,000–1,700**	**SUPPORTED ESTIMATE**	A model-development range, not a declaration of canon.

The range is intentionally broad because the two most consequential variables—J2 occupancy and the physical scale of ARU/Red Wash—are unresolved. It should not be narrowed by guessing.

There is also a structural ratio problem. With all 237 J2 billets occupied:

$$\frac{237}{708} = 33.5\%$$

of PR #9's current enterprise baseline. At 15% of enterprise population, 237 billets imply a company of **1,580** employees; at 10%, they imply **2,370**. That does not prove either ratio is the “correct” one. It demonstrates that the J2 decision has enterprise-scale financial consequences and cannot be appended to the old company model as another small headquarters department. The J2 charter itself emphasizes intentionally

deep professional capacity, research resources, education, travel, field support, fellowships, sabbaticals, and first-class administrative/workplace support.

Compensation basis. March 2026 BLS data provide useful floors and midpoints: transportation/warehousing employer compensation was approximately $49.04 per hour, management/business/financial occupations about $72.24, professional/scientific/technical services $75.15, and information $83.81. [1] Annualized at 2,080 hours purely for scale comparison, those figures are approximately $102,000, $150,000, $156,000, and $174,000 of employer compensation respectively. Sable Harbor's own doctrine calls for strong base pay, meaningful long-duration participation, learning time as paid work, substantial development, hybrid-work infrastructure, and employee continuity. A finance model should therefore add role-specific equipment, travel, facilities, recruiting, training, security, and technology costs rather than treating base salary plus payroll tax as “fully loaded.”

A first bottom-up headquarters architecture should look approximately as follows:

Corporate institution	Planning FTE	Annual recurring cash-cost envelope	Major drivers	State
CEO/executive office, Board, Corporate Secretary	8–15	$3–6M	senior compensation, board fees, governance, travel, records	**SUPPORTED ESTIMATE**
Finance	24–40	$5–9M	accounting, FP&A, treasury, tax, consolidation, controls	**SUPPORTED ESTIMATE**
Legal, Risk & Compliance, Internal Audit	25–45	$6–11M	legal labor, outside counsel, compliance, audit, insurance interfaces	**SUPPORTED ESTIMATE**
People & Culture	20–35	$4–7M	talent, employee relations, benefits administration, mobility, management formation	**SUPPORTED ESTIMATE**
Enterprise Technology Services	40–70	$12–25M	labor, enterprise SaaS, compute, developer platform, service engineering	**SUPPORTED ESTIMATE**
Enterprise Security / CISO	20–35	$7–14M	security engineering, identity, monitoring, tooling, testing, third parties	**SUPPORTED ESTIMATE**

Corporate institution	Planning FTE	Annual recurring cash-cost envelope	Major drivers	State
Procurement, safety, quality, workplace/ facilities and ESS coordination	25–45	$8–16M	procurement, facilities, standards, safety/ environment, workplace services	**SUPPORTED ESTIMATE**
Non-J2 corporate subtotal	**162–285**	**$45–88M**		**SUPPORTED ESTIMATE**
J2	**170–237 occupied**	**$45–70M**	professional labor, education, travel, research, field collection, facilities/support	**SCENARIO**
Central institutional total	**332–522**	**$90–158M**		**SUPPORTED ESTIMATE / SCENARIO**

The organizational logic is directly supported by current doctrine. ESS now includes OGC, Risk & Compliance, Finance, People & Culture, Technology Services, enterprise security, procurement/vendor support, enterprise safety/environmental governance where justified, quality/technical stewardship, facilities/workplace, and continuity coordination; Internal Audit remains separately independent. Technology Services is itself defined as an internal product/engineering organization spanning employee technology, enterprise platforms, developer systems, cloud/compute, enterprise AI, architecture/supply chain, and reliability engineering.

The implication is stark: **at PR #9's $178.6 million revenue, a $90–158 million central institutional burden would consume roughly 50–88% of revenue before the direct costs of building Foundry, running Red Wash, operating ARU, staffing Atlas, funding Willow, or serving customers.** That alone is enough to reject $178.6 million as the natural steady-state scale of the current company architecture.

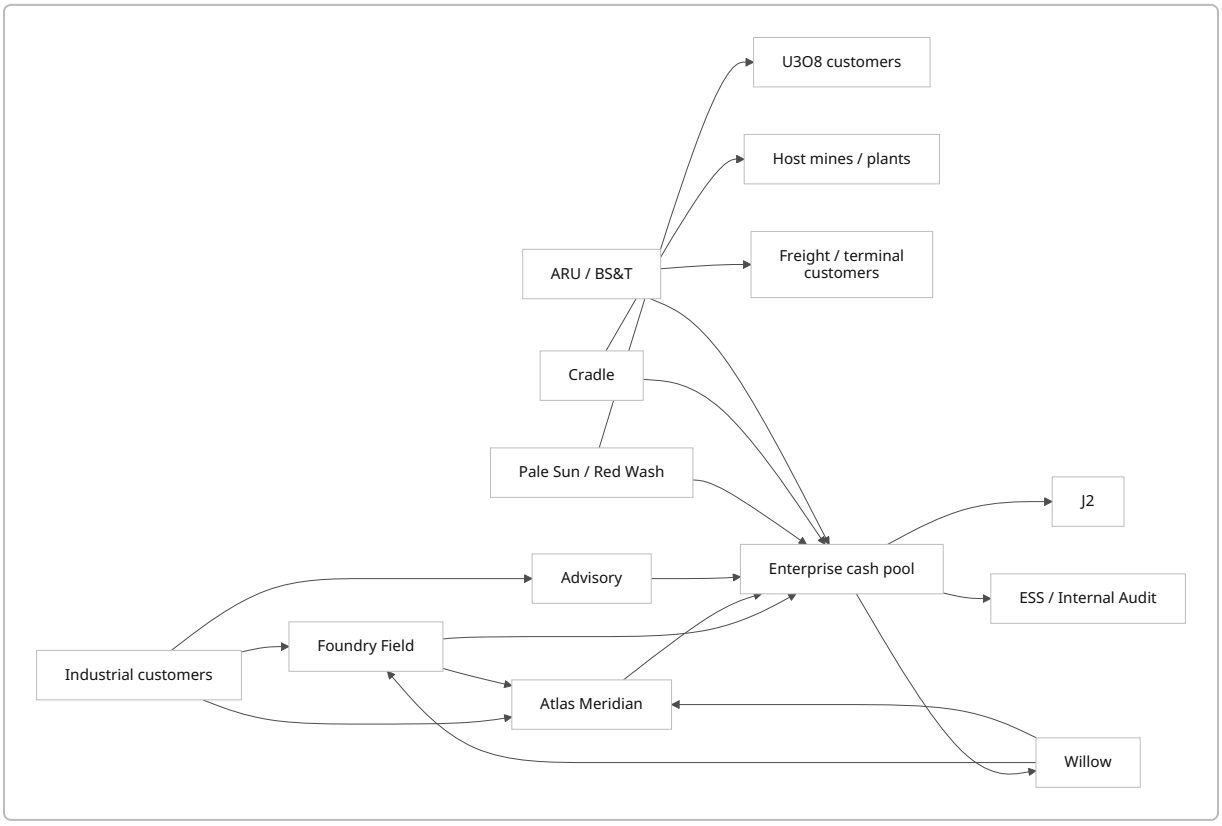
The 47,000-person figure fails a similar but much larger test. Using a scenario range of $150,000–$225,000 per employee for employer compensation plus ordinary role support gives:

47,000-person scenario	Lower	Mid	Upper
Cost per employee	$150k	$185k	$225k
Annual workforce cost	$7.05B	$8.70B	$10.58B
Revenue if workforce cost = 50% of revenue	$14.10B	$17.39B	$21.15B
Revenue if workforce cost = 40% of revenue	$17.63B	$21.74B	$26.44B

Those are scenario mechanics, not forecasts. Their purpose is to show the order of magnitude. A 47,000-person Sable Harbor would be a **multi-tens-of-billions industrial enterprise**, not the company described by the current canon.

Revenue, cost and capital mechanics

The portfolio is economically heterogeneous by design. Current canon describes Sable Harbor not simply as mining software but as an industrial-systems company in which Foundry represents, Willow tests, Atlas investigates, Pale Sun owns and operates, Cradle recovers, ARU moves, and Advisory transfers the method. Financial modeling should therefore reject a single SaaS gross-margin assumption across the group.



Revenue and cost architecture by business

Business	Revenue mechanism	Direct cost mechanism	Capital / balance-sheet mechanism	Current evidence state
Foundry Field	recurring platform access, usage/ subscription structures, implementation/ deployment, support and specialist work; exact packaging/ pricing OPEN	product engineering, customer delivery, support/SRE, cloud, integration, field travel, third-party software	capitalized/ internal-use software policy questions, receivables, deferred revenue, implementation WIP where appropriate	Mature commercial behavior **LOCKED**; quantitative economics **OPEN**
Atlas Meridian	controlled commercial licenses/ design partnerships/ analytical engagements; exact launch and pricing OPEN	AI/model compute, data, science/ engineering, validation, domain experts, customer work	model/data infrastructure, possible contract liabilities, compute commitments	Controlled commercialization **LOCKED direction**; economics **OPEN**
Willow	primarily institutional R&D; occasional customer-funded work possible but no standalone revenue doctrine locked	research engineering, prototypes, machine shop/lab, sensors, travel, destructive testing, field qualification	lab equipment, prototypes, experimental assets; R&D capitalization policy must be conservative	Institutional role **LOCKED**; revenue **OPEN**
Pale Sun / Red Wash	sale of uranium concentrate; no enrichment or fuel fabrication	mining, development, ground support, maintenance, milling, labor, reagents, energy, environmental/ safety, transport	mineral interests, mine development, plant/equipment, inventory, ARO/ closure, reclamation, acquisition accounting	Operating model **LOCKED**; production/ economics **OPEN**
Cradle	equipment/service economics, recovery participation, stream-specific rights, royalty-like structures or combinations	metallurgy, pilot work, sampling, field labor, process engineering	pilot/recovery equipment, IP/ process rights, possible host receivables/ investments	Mechanism **LOCKED**; customer/ material/economics **OPEN**

Business	Revenue mechanism	Direct cost mechanism	Capital / balance-sheet mechanism	Current evidence state
ARU / BS&T	freight, terminal, storage, handling and related logistics	crews, fuel, maintenance, track/terminal expense, leased/owned rolling stock, dispatch, safety	rolling stock, track/terminal assets, leases, working capital, acquisition accounting	Operating role **LOCKED**; footprint/economics **OPEN**
Advisory	professional-fee engagements transferring Sable Harbor's operating/analytical method	expert labor, travel, subcontractors, engagement support	working capital/WIP, limited fixed capital	Emergence **LOCKED direction**; structure/pricing/P&L **OPEN**
J2	no external revenue required by canon	specialized labor, education, research, collection, travel, field support, facilities	primarily institutional operating cost and selected equipment	237-billet establishment **LOCKED**

Foundry's foundational economics are qualitatively well supported: customers began to value continuing access to maintained mappings, relationships, provenance and decision context rather than only one-time consulting output, and the 2021 threshold was repeatable deployment without custom code or founder heroics. That is a credible path toward a high-margin recurring product. External analogs show mature industrial software can indeed generate 80%+ gross margins, but Sable Harbor's continuing field/delivery intensity argues for a lower blended Foundry margin than a pure subscription vendor. [7]

Atlas is different. It is explicitly a controlled 2026 commercialization of an investigative system rather than an autonomous decision engine. Its earliest economics should therefore tolerate substantial science, evaluation, compute and customer-validation expense rather than forcing mature-software margins immediately.

Red Wash is the largest statement-readiness blocker. Canon says it is an underground Wyoming uranium mine with selective mining, surface stockpiles, crushing/milling, and uranium-concentrate output, while explicitly leaving reserves, mine plan, workforce, permits, production, closure obligations, and economics open. That means the mine cannot have a defendable P&L or balance sheet until at minimum the following chain exists:

$$\text{tonnes mined} \times \text{grade} \times \text{recovery} \rightarrow \text{lbs } U_3O_8 \text{ produced} \rightarrow \text{inventory} \rightarrow \text{lbs sold} \times \text{realized price}$$

followed by mining/milling costs, depreciation/depletion, sustaining capital, closure/ARO, reclamation, freight, and working capital. Current uranium-company filings show why the distinction matters: extracted pounds, produced concentrate, contracted deliveries and revenue recognition can occur in different periods. [8]

ARU needs an analogous physical driver model:

$$\text{carloads / tons} \times \text{distance or movement} \times \text{rate} + \text{terminal/storage/handling} = \text{external revenue}$$

with crew, fuel, maintenance, locomotive/car leases or depreciation, track/terminal maintenance, insurance and safety expense beneath it. PR #9's $23.6 million revenue assumption is possible for a small operation, but it would fall below the STB's 2025 $49.1 million Class II threshold, making it a Class III-scale railroad on revenue if BS&T alone were assessed on that basis. [4] Canon has not yet decided whether that is the desired scale.

Existing quantitative financial anchors

Anchor	Value	State	Financial treatment
The Crossing runway	~12 weeks	**LOCKED**	Historical liquidity constraint; not a complete 2020 cash-flow statement.
Evalon “tractor parts” expense	$4,812.73	**LOCKED**	Genuine historical expense/lore anchor; immaterial to enterprise forecasting but useful for synthetic ledger continuity.
Harrison Vale 2021 financing	~ $48M	**LOCKED**	Equity-financing anchor.
Wolf Ridge 2022 financing	~ $135M	**LOCKED**	Equity-financing anchor.
2022 post-money valuation	~ $900M	**LOCKED**	Financing valuation anchor, not current valuation.
Post-2022 ownership working split	~66.5% insiders / 33.5% outside	**LOCKED working record**	Cap-table constraint requiring later share-level reconstruction.
Founder secondary in 2022	none	**LOCKED**	Financing cash went to company rather than founder liquidity in that round.
Red Wash acquisition	closed around July 2025	**LOCKED spine**	Price, financing, liabilities and purchase accounting remain OPEN.
ARU acquisition	acquisition is LOCKED	**LOCKED spine**	Price, assets, debt, routes, workforce and transaction terms OPEN.
426 FTE / 34 contractors	old 2026 model	**SUPERSEDED / legacy calibration**	May inform sensitivity analysis; cannot be current census.

Anchor	Value	State	Financial treatment
$124.5M revenue / 65% GM / $(13.3)M operating loss	old 2026 model	**SUPERSEDED / legacy calibration**	Historical model input only.
$172.5M equity raised / $25M facility / $10M drawn	old model	**CONFLICT / SUPERSEDED calibration**	Must be rebuilt from current financing canon.
PR #9 708 employees / 61 contractors	working finance model	**PROVISIONAL ASSUMPTION**	Not company truth.
PR #9 FY2026 revenue $178.6M	working finance model	**PROVISIONAL ASSUMPTION**	Underscaled against current institutional architecture.
PR #9 Red Wash $24.7M rev / $22.3M op cost	working finance model	**SCENARIO**	Canon leaves mine economics open.
PR #9 ARU $23.6M rev / $20.3M op cost	working finance model	**SCENARIO**	Canon leaves logistics economics open.
PR #9 $48M Red Wash EV / $76M ARU EV / $18.5M ARO	working finance model	**SCENARIO**	No present canon support.
PR #9 ~$70M incremental equity + ~$86M acquisition debt	working finance model	**PROVISIONAL ASSUMPTION**	Acquisition financing remains OPEN.

The financing conflict is important. The old $172.5 million lifetime-equity calibration cannot simply survive unchanged because current governance locks approximately $183 million in the 2021 and 2022 rounds alone. That does not prove a new lifetime-total number—the word “approximately” matters and earlier/other capital remains to be reconstructed—but it proves the old figure requires retirement or explicit reconciliation.

Preliminary reverse-solved enterprise requirement

The following is the appropriate order-of-magnitude financial envelope for model-building, not a forecast:

Economic state	Revenue	Consolidated gross margin	EBITDA	Operating cash flow	Sustaining/ growth capex	Free cash flow	Interpretation
Minimum sustainable	$450–600M	48–55%	3–8%	$20–50M	$30–55M	negative to ~$20M	Can carry institutions only with restrained J2 occupancy, modest industrial capex and continued prioritization.
Healthy	**$650–900M**	**52–60%**	**10–15%**	**$65–125M**	**$40–70M**	**$25–70M**	Best preliminary target zone for current architecture.
High-performance	$900M–$1.2B	55–62%	15–20%	$130–210M	$55–90M	$70–140M	Supports strong reinvestment, industrial resilience and optionality without SaaS-like assumptions across the whole enterprise.

All of these ranges are **SUPPORTED ESTIMATE / SCENARIO**. The healthy-case logic is that roughly $90–158 million of central institutional capacity should not consume half of company revenue. At approximately $700 million, a $120 million central burden is about 17% of revenue; at $178.6 million it is about 67%. That is the financial reason the existing PR #9 scale should be recalibrated.

PR #9 and statement readiness

PR #9 deserves a nuanced disposition. **Its engineering work is valuable; its company model is not authoritative.** The PR records explicit run identity, migration discipline, scenario coexistence, lineage, accounting modules, reporting queries, workbooks, release controls and database validation. It also openly states that its 2016–2022 revenue values are only calibration anchors and that the PR is review-blocked and must not be merged or described as production-ready. Those are good controls.

Its quantitative assumptions are similarly transparent. The PR's own reconciliation says Alternative B is merely `MODEL_PROPOSED`, reversible, and contains no locked numbers; its assumption register identifies the architecture as model-proposed. The failure is not covert fabrication. The failure is that **the modeled company predates and undershoots the institutional architecture now locked on `main`.**

The principal reconciliation defects are:

Finding	Assessment	State
PR #9 remains unaccepted and review-blocked	Explicit in PR and repository issue state.	**LOCKED repository state**
708-person census does not reconcile J2	J2's 237-billet establishment is now locked and is not visibly allocated within the 708 baseline.	**CONFLICT**
426/431-person CoreCo anchor comes from a historical branch	Current README says old operating-model branch is noncontrolling and exact total headcount remains open.	**SUPERSEDED calibration**
2026 Emberline treatment needs lineage-only handling	Current lore says Emberline is active through 2025 and its enduring work is absorbed; it is not a standalone 2026 core division.	**CONFLICT** where old staffing persists
Old $172.5M funding anchor conflicts with newer locked financing facts	2021 + 2022 rounds alone are approximately $183M.	**CONFLICT**
Federal/legal-entity assumptions are not settled canon	Detailed entity/jurisdiction mechanics remain open on current main.	**OPEN**
Red Wash workforce, revenue, operating cost, purchase price and ARO are assumed	Current canon expressly leaves physical/economic model and acquisition terms open.	**PROVISIONAL ASSUMPTION / SCENARIO**
ARU workforce, revenue and purchase price are assumed	Current canon leaves footprint, workforce, assets, routes and transaction terms open.	**PROVISIONAL ASSUMPTION / SCENARIO**
$178.6M enterprise revenue is institutionally underscaled	At current central-capability requirements it leaves insufficient economic room for business direct cost.	**DERIVED conflict**
Accounting/platform implementation has reusable value	Migration, run-identity, lineage, ledger, operating-flow and reconciliation architecture need not be discarded merely because assumptions change.	**SUPPORTED retention**

The correct technical response is therefore **not “delete PR #9.”** It is:

Keep the accounting and data-platform machinery; invalidate or version-separate the obsolete company assumptions; rebuild the quantitative baseline against current `main`; then rerun its validation framework.

Financial-statement readiness

Area	Readiness	Why
Historical 2016–2020 P&L	Low	Services-first history and Crossing liquidity are known qualitatively, but no complete revenue/payroll/vendor/ledger spine is canonical.
Historical 2021–2022 financing	Medium	Major equity rounds and 2022 valuation are locked, but complete capitalization/share accounting and historical statements are not.
2026 Foundry Field P&L	Medium-low	Commercial product behavior is mature; contract counts, pricing, ARR, customer concentration and direct-cost allocation remain open.
2026 Atlas P&L	Low	Commercialization stage is known, but customer/pricing/ compute economics are open.
2026 Willow cost-center statement	Medium	Institutional purpose and activities are clear; headcount, facilities and research budget are missing.
2026 Red Wash P&L	Very low	No canonical production, grade, recovery, sales, workforce, mine plan or cost curve.
Red Wash balance sheet	Very low	Purchase price allocation, mineral value, PP&E, inventory opening balance, debt, ARO and closure liability are unresolved.
2026 ARU/BS&T P&L	Very low	Routes, traffic, customers, pricing, fleet, terminals, workforce and costs are open.
ARU balance sheet	Very low	Purchase price, assets, leases/debt, working capital and PPA are open.
Cradle P&L	Low	Commercial mechanism known; exact stream/customer/ economics open.
Advisory P&L	Low	Practice concept known; headcount, pricing, utilization, leader and formal structure open.
Corporate / ESS budget	Medium-low	Functional architecture is unusually well specified; staffing and dollars are not.
J2 budget	Medium	Billet establishment and institutional model are unusually strong; occupancy, compensation bands and program budget are open.

Area	Readiness	Why
Consolidated 2026 income statement	**Low**	Too many material segment drivers remain open.
Consolidated balance sheet	**Very low**	Acquisition accounting, debt, cash, PP&E, inventory, ARO and legal entities remain unresolved.
Consolidated cash-flow statement	**Very low**	Cannot be credible before opening balance sheet, acquisitions, capex, debt and working-capital drivers exist.

Accordingly, **no final consolidated statements should yet be generated and presented as company truth**. A model may generate scenarios, but every output should inherit the state of its inputs.

Implications, limitations and decision queue

The larger financial implication is that Sable Harbor's September 3 canon has transformed the company from a modest venture-backed mining-software company into a deliberately complex hybrid institution. The enterprise now combines software economics, AI/research economics, professional-services economics, mine economics, logistics economics, and a large non-revenue institutional-intelligence capability. Current headquarters doctrine makes that complexity intentional: operating businesses retain real end-to-end authority, ESS supports rather than commands them, J2 remains institutionally separate, Internal Audit is independent, and horizontal professional practices do not become shadow operating chains.

That architecture can be financially coherent, but only at greater scale than the old model. **The company does not need to become a 47,000-person giant; it does need either materially more revenue, materially lower institutional occupancy, or both.** The current healthy planning zone of roughly $650–900 million revenue is not a lore claim. It is the preliminary level at which the institutional structure stops dominating the economics and the physical businesses can be maintained without assuming miraculous margins.

The financing history also needs to tell a believable bridge from the approximately $900 million 2022 valuation to the 2025 acquisition era. Current canon establishes real institutional capital and balanced governance but intentionally leaves Red Wash and ARU purchase prices and financing open. That is preferable to retaining PR #9's arbitrary approximately $70 million equity plus $86 million debt acquisition package as if it had already happened.

The physical businesses should be solved **from operating reality forward**, not from a desired consolidated statement backward. Red Wash needs a mine/mill/uranium-sales driver model; ARU needs traffic, route, fleet, terminal and rate drivers; Cradle needs a contract and process-economics model. Industry evidence shows why this matters: uranium sales can lag or differ from production because inventory and contract timing matter, while railroad revenue classification itself changes materially around the STB's ~$49.1 million Class II threshold. [9]

The main limitations of this report are deliberate. Exact occupied J2 headcount is unknown. Exact enterprise headcount is open. Red Wash and ARU physical scopes are open. No customer contract registry or complete 2026 payroll exists as canon. Acquisition prices, cash/debt balances and current cap table are unresolved.

The proposed workforce, cost and revenue ranges above therefore **must remain model inputs** until controlled decisions replace them. Current canon explicitly requires that these matters not be silently invented.

The shortest high-value decision queue is:

Priority	Decision required	Why it unlocks the model	Recommended evidence state after decision
Immediate	**Set J2 2026 actual/average occupancy against the 237-billet establishment**	Largest unresolved institutional cost and headcount variable; determines total-company scale.	**LOCKED** or clearly dated **PROVISIONAL ASSUMPTION**
Immediate	**Choose the enterprise workforce scale architecture**: approximately 1,000–1,300 lean, 1,400–1,700 mature, or a separately justified alternative	Prevents every payroll, benefits, facilities and technology line from being a plug.	**PROVISIONAL ASSUMPTION**
Immediate	**Define Red Wash operating case**: mine/mill workforce, tonnes, grade, recovery, production, sales schedule, realized-price method, opex, sustaining capex and ARO	Unlocks the mine P&L, inventory, PP&E, ARO, cash flow and acquisition economics.	**PROVISIONAL ASSUMPTION / SCENARIO**, then canon lock
Immediate	**Define ARU/BS&T physical scale**: route/terminal footprint, fleet, traffic, workforce and external-customer revenue class	Unlocks logistics P&L, fixed assets, maintenance capex and acquisition value.	**PROVISIONAL ASSUMPTION / SCENARIO**
Next	**Lock 2021–2026 capitalization and acquisition financing bridge**	Replaces the obsolete $172.5M / $25M-debt calibration and creates a real balance-sheet opening position.	**LOCKED**
Next	**Set Foundry commercial driver set**: customers, ARR/recurring revenue, implementation volume, pricing and renewal assumptions	Gives the enterprise a credible cash-generating core rather than a consolidated revenue plug.	**PROVISIONAL ASSUMPTION**

Priority	Decision required	Why it unlocks the model	Recommended evidence state after decision
Next	**Set Atlas, Cradle and Advisory 2026 commercialization scale**	Prevents emerging businesses from becoming unexplained revenue or expense residuals.	**PROVISIONAL ASSUMPTION**
Then	**Build the corporate/J2 cost budget from actual roles and policies**	Makes ESS/J2 a governed institutional investment rather than SG&A percentage-of-revenue.	**DERIVED** from census/ compensation decisions
Then	**Rebase PR #9 inputs and rerun its ledger/reconciliation stack**	Preserves valuable engineering while replacing invalid company assumptions.	Generated outputs inherit input states
Final gate	**Publish controlled 2026 opening balance sheet, monthly P&L/cash flow, business-line P&Ls and consolidated statements in triplicate**	Only after reconciliation proves statements arise from business drivers rather than plugs.	**LOCKED / DERIVED** as appropriate

The governing financial principle should mirror the company's own lore: **do not force the numbers to converge merely because a spreadsheet wants one answer. Preserve what is known, what is assumed, what is still open, and the causal path from operating reality to financial statement.** Foundry's original lesson was that reconciliation does not necessarily mean convergence, and the company now needs to apply that same discipline to itself.

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1 Employer costs per hour worked for employee compensation of private industry workers, by industry
https://www.bls.gov/charts/employer-costs-for-employee-compensation/costs-by-industry.htm?utm_source=chatgpt.com

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https://www.sec.gov/Archives/edgar/data/1031308/000103130826000007/bsy-20251231.htm

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4 Surface Transportation Board
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